

Module 2: Technologies & Business Models

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Relationship Between E-Commerce & Networking

Networking refers to the process of connecting computers and devices through communication channels to share data, resources, and information. It can be done using wired or wireless technologies. In simple terms, networking enables multiple systems to communicate and exchange information efficiently.

Role of Networking in E-Commerce

Networking plays a fundamental role in the functioning of e-commerce systems.

- It enables communication between buyers, sellers, and service providers
- It supports online transactions and payment processing
- It allows real-time data transfer (orders, inventory updates, tracking)
- It connects different components like websites, servers, and databases
- It facilitates customer interaction through emails, chats, and support systems

Without networking, e-commerce platforms cannot operate.

How E-Commerce Depends on Networking

E-commerce is completely dependent on networking infrastructure for its operation.

- Online stores require internet connectivity to be accessible
- Payment gateways depend on secure networks to process transactions
- Order processing and tracking require continuous data exchange
- Cloud storage and databases rely on network connectivity
- Communication between businesses and customers happens through networks

Thus, networking forms the backbone of all e-commerce activities.

Benefits of Networking in E-Commerce

- Enables fast and efficient communication
- Supports global connectivity and market reach
- Allows real-time updates and transaction processing
- Improves coordination between business operations
- Enhances customer experience through quick responses
- Reduces operational time and cost

Types of E-Commerce/Networking Commerce

1. Models Based on Transaction Party

2. B2C (Business to Consumer): This refers to transactions where businesses sell products or services directly to end consumers. In this model, companies create online platforms (websites/apps) where customers can browse, select, and purchase products.

- Large customer base
- Fast transactions
- Use of digital marketing
- Fixed pricing (generally)

Examples:

- Online retail stores
- Subscription service

3. B2B (Business to Business): This involves transactions between two businesses or companies. In this model, one business sells goods or services to another business, such as manufacturers selling to wholesalers or wholesalers selling to retailers.

- Bulk transactions
- Long-term relationships
- Negotiated pricing
- Complex supply chain

Examples:

- Wholesale platforms
- Supplier portals

4. C2C (Consumer to Consumer) This refers to transactions between individual consumers. In this model, individuals sell products or services to other individuals through online platforms that act as intermediaries.

- Platform-based selling
- Lower prices
- Used goods or second-hand items
- Direct interaction between users

Examples:

- Online resale platforms
- Auction websites

5. C2B (Consumer to Business): This involves individuals offering products or services to businesses. In this model, consumers create value that businesses purchase, such as freelance services, content creation, or product feedback.

- Reverse business model
- Individuals have pricing power (sometimes)
- Service-oriented

Examples:

- Freelancing platforms
- Influencer marketing

2. Models Based on Transaction Type

These models define **how value is created, delivered, and revenue is generated** in e-commerce systems.

1) Brokerage Model: This model acts as an intermediary between buyers and sellers and earns a commission for facilitating transactions. **Examples:** Uber, Airbnb, online marketplaces

- Platform connects buyers and sellers directly
- Charges commission or service fee per transaction
- Does not own goods or services

2) Advertising Model: This model provides free or low-cost services and earns revenue through advertisements. **Examples:** Google Search, Meta (Facebook, Instagram)

- Focus on attracting high user traffic
- Advertisers pay for visibility and targeted ads
- Revenue depends on user engagement

3) Infomediary Model: This model collects, analyzes, and sells user data or market information to businesses. **Examples:** Market analytics platforms, data-driven ad networks

- Data is the main source of value
- Tracks user behavior and preferences
- Helps businesses in targeted marketing

4) Merchant Model: This is a traditional online retail model where the business buys and sells products at a profit. **Examples:** Amazon (retail segment), Flipkart

- Company owns inventory
- Sells products at marked-up prices
- Handles storage, delivery, and sales

5) Subscription Model: In this model, users pay a recurring fee to access services or content. **Examples:** Netflix, Spotify, SaaS platforms

- Monthly or yearly payments
- Continuous access to services
- Focus on customer retention

3. E-Governance Model

E-Governance refers to the use of electronic systems by the government to deliver services efficiently to citizens and businesses.

1) G2C (Government to Consumer): This involves government providing services directly to citizens through online platforms. Governments use e-commerce systems to deliver public services efficiently and transparently.

- Easy access to services
- Reduces paperwork
- Saves time

Examples:

- Online tax filing
- License applications

2) G2B (Government to Business): This involves transactions between government and businesses. Businesses interact with government portals for regulatory compliance, tax payments, and licenses.

- Transparency
- Faster approvals

- Digital documentation

Examples:

- GST filing
- Online tenders

3) G2G (Government to Government): This involves transactions and communication between different government departments or agencies through online systems. It is used to share information and coordinate activities within government bodies for better administration and efficiency.

- Improves coordination between departments
- Enables fast and secure data sharing
- Reduces delays in government processes
- Increases efficiency in administration

Examples:

- Data sharing between central and state governments
- Communication between ministries and departments
- Integrated government information systems

Advantages of E-Governance

- Reduces paperwork and manual processes
- Improves transparency and accountability
- Faster delivery of services
- Reduces corruption and delays

Internet, Intranet & Extranet

1) Internet: The Internet is a global network of interconnected computers that communicate using standard protocols to share information worldwide.

- Worldwide connectivity
- Open and public network
- Supports multiple services (websites, email, file transfer)
- Uses standard protocols like HTTP, TCP/IP
- Accessible to anyone with internet connection

Uses in E-Commerce

- Hosting online stores and websites
- Enabling online transactions and payments
- Digital marketing and advertising
- Customer interaction through emails and chat
- Order placement and tracking

2) Intranet: An Intranet is a private network used within an organization that uses internet technologies but is accessible only to authorized users.

- Restricted access (internal use only)
- Secure and controlled environment
- Faster communication within organization
- Uses same technologies as internet (web browsers, servers)

Uses in Organizations

- Sharing internal information and documents
- Employee communication
- Managing company data and resources
- Internal applications like HR, payroll systems

3) Extranet: An Extranet is a private network that allows limited access to external users such as suppliers, customers, or partners.

- Controlled external access
- Secure data sharing between organizations

- Extends intranet to outside users
- Requires authentication (login, password)

Uses in Business

- Collaboration with suppliers and partners

- Sharing order details and inventory data
- Business-to-business communication
- Supply chain management

Basis	Internet	Intranet	Extranet
Access	Public	Private (within organization)	Limited external access
Users	Anyone	Employees only	Employees + authorized outsiders
Security	Less secure	Highly secure	Moderately secure
Purpose	Information sharing globally	Internal communication	External collaboration
Ownership	No single owner	Owned by organization	Shared between organizations

EDI Systems (Electronic Data Interchange)

Electronic Data Interchange (EDI) refers to the electronic exchange of business documents between organizations in a standardized format without human intervention. It replaces traditional paper-based documents like invoices, purchase orders, and shipping notices with digital communication. EDI is a crucial technology in e-commerce that automates the exchange of business documents. It improves speed, accuracy, and efficiency but requires proper infrastructure and standardization

Working of EDI

The working of EDI involves the following steps:

1. A business generates a document (e.g., purchase order) in its internal system.
2. The document is converted into a standard EDI format using EDI software.
3. The formatted document is transmitted through a communication network.
4. The receiving system translates the EDI message into a readable format.
5. The document is processed automatically in the receiver's system.

This process eliminates manual handling and speeds up transactions.

Advantages of EDI

- Reduces paperwork and manual processing
- Speeds up business transactions

Components of EDI

1. **EDI Standards:** These define the format of electronic documents (e.g., structure and data fields) to ensure compatibility between systems.
 2. **EDI Software:** This is used to convert internal data into EDI format and vice versa.
 3. **Communication Network:** This enables the transfer of EDI documents between organizations (internet or private networks).
 4. **Business Applications:** These are internal systems (like ERP) that generate and process business documents.
- Improves accuracy by reducing human errors
 - Enhances business efficiency
 - Reduces operational costs

- Enables better coordination between organizations

Limitations of EDI

- High initial setup cost
- Requires standardization between partners

- Limited flexibility in document formats
- Security concerns during data transmission
- Requires technical expertise

Wireless Application Protocol (WAP)

Wireless Application Protocol (WAP) is a set of communication protocols that allows mobile devices to access internet services and information over wireless networks. It was designed to enable basic web access on early mobile devices with limited processing power and small screens.

WAP was an important early technology that enabled mobile internet access. Although it is now mostly replaced by modern smartphones and high-speed networks, it played a key role in the development of mobile commerce.

- Provides internet access on mobile devices
- Works on wireless communication networks
- Supports lightweight web pages (WAP pages)
- Optimized for low bandwidth and small screens
- Enables basic services like browsing, email, and messaging

Working of WAP

The WAP system works through the following process:

1. The user sends a request from a mobile device.
2. The request goes to a WAP gateway server.
3. The gateway converts the request into a standard web format (HTTP).
4. The web server processes the request and sends back the response.
5. The WAP gateway converts the response into a mobile-friendly format.
6. The content is displayed on the mobile device.

- Sending and receiving emails
- Mobile banking services
- News and weather updates
- Simple online transactions

Advantages of WAP

- Enables mobile internet access
- Works on low-speed networks
- Requires less data and bandwidth
- Accessible on basic mobile devices

Limitations of WAP

- Limited graphics and functionality
- Slow performance compared to modern internet
- Poor user experience
- Largely outdated due to advanced mobile technologies

Uses of WAP

- Accessing basic websites on mobile phones

Mobile Technologies in E-Commerce

- 1) **Hand Held Devices:** Hand held devices are small, portable electronic devices that can be easily carried and used by individuals. These devices allow users to access digital information and services anytime and anywhere.

Types

1. **Smartphones:** These are advanced mobile phones with internet access, apps, and communication features. They are widely used for online shopping, banking, and communication.
2. **Tablets:** These are larger than smartphones with bigger screens. They are used for browsing, reading, and online transactions.
3. **PDA's (Personal Digital Assistants):** These are early portable devices used for storing data, scheduling, and basic

communication. They are now mostly replaced by smartphones.

Role in E-Commerce

- Enable mobile shopping and payments
- Provide access to e-commerce apps and websites
- Support digital wallets and online banking
- Improve customer convenience and accessibility
- Allow businesses to reach customers anytime

- 2) **Mobility & Commerce (M-Commerce):** Mobility refers to the ability to access information and services from anywhere using portable devices and wireless networks. Mobile Commerce (M-Commerce) refers to buying and selling goods and services through mobile devices such as smartphones and tablets.

- Wireless connectivity
- Location-based services
- Portability and convenience
- Instant transactions
- Personalized user experience
- Supports mobile payments
- Enhances customer engagement

Limitations of M-Commerce

- Security concerns
- Small screen size limitations
- Dependence on network connectivity
- Limited battery life
- Device compatibility issues

Advantages of M-Commerce

- Easy and convenient access
- Faster transactions
- Wide customer reach

- 3) **Mobile Computing:** Mobile computing refers to the use of portable devices and wireless networks to access and process data while moving from one place to another.

Components of Mobile Computing

- **Devices:** Includes smartphones, tablets, and laptops used for accessing data.

- **Networks:** Wireless communication systems such as Wi-Fi and mobile data networks.
- **Software:** Applications and operating systems that enable mobile functionality.

Applications

- Online shopping and banking

- Email and communication services
- GPS and navigation systems
- Social media and entertainment
- Business and enterprise applications

Wireless Web

Wireless Web refers to accessing internet services and websites using wireless devices such as smartphones, tablets, and laptops without the need for wired connections.

It allows users to browse, communicate, and perform online transactions through wireless networks like Wi-Fi and mobile data.

- Provides internet access without physical cables
- Supports mobility and access from any location
- Compatible with mobile devices and handheld systems
- Uses wireless technologies like Wi-Fi, 4G, and 5G
- Enables real-time communication and data transfer

Advantages

- High convenience and portability
- Easy access to information anytime and anywhere
- Supports mobile commerce and online services
- Reduces dependency on fixed infrastructure
- Enables quick communication and transactions

Web Security

Web security is essential to protect websites, users, and data from cyber threats and unauthorized access.

- Protects sensitive information like passwords and payment details
- Ensures safe online transactions
- Prevents data breaches and cyber attacks
- Maintains trust between users and businesses
- Protects website integrity and performance

Security Measures

1. **Encryption:** Converts data into a secure format to prevent unauthorized access during transmission.
2. **SSL (Secure Sockets Layer):** Provides secure communication between web browser and server by encrypting data.
3. **Firewalls:** Security systems that monitor and control incoming and outgoing network traffic.

Infrastructure Requirement for E-Commerce

E-commerce requires a strong and reliable infrastructure to operate efficiently. This infrastructure includes hardware, software, networking, security systems, and human resources that work together to support online business activities.

- 1) **Hardware Requirements:** refers to the physical components needed to run e-commerce systems.

- Servers for hosting websites and databases

- Computers and mobile devices for users and administrators
 - Storage devices for data management
 - Backup systems for data recovery
 - Networking devices like routers and switches
- 2) **Software Requirements:** includes programs and applications required for managing e-commerce operations.
- Web server software to host websites
 - Database management systems (DBMS)
 - E-commerce platforms and applications
 - Operating systems
 - Customer relationship management (CRM) software
- 3) **Network Requirements:** A reliable network is essential for communication and data transfer.
- High-speed internet connectivity
 - Secure communication channels
 - Network protocols for data exchange
 - Connectivity between servers, users, and payment systems
- 4) **Human Resources:** Skilled personnel are required to manage and maintain the e-commerce system.
- IT professionals (developers, network administrators)
 - Customer support staff
 - Digital marketing experts
 - Security specialists
 - Management team
- 5) **Payment Systems:** enable online financial transactions.
- Payment gateways for processing payments
 - Digital payment methods (cards, UPI, net banking)
 - Integration with banks and financial institutions
 - Secure transaction processing
- 6) **Security Infrastructure:** protects the system from cyber threats.
- Encryption technologies
 - Secure protocols like SSL
 - Firewalls and antivirus systems
 - Authentication and access control mechanisms
 - Regular security updates and monitoring